18B1WCI674: MACHINE LEARNING LAB

Assignment-5 (Scikit-learn: Machine Learning Modeling)

Jan 05, 2026

Instructions

1. Use Jupyter Notebook or Google Colab for this assignment.
2. Load the dataset **titanic.csv** file using Pandas.
3. Use `scikit-learn` for preprocessing, model training, and evaluation.
4. Follow the complete machine learning workflow.
5. Avoid data leakage during preprocessing.
6. Clearly write observations after each major step.
7. Submit the notebook and dataset within lab hours.

Overview of Scikit-learn

Scikit-learn is a powerful Python library used for implementing ***machine learning algorithms, preprocessing techniques, and model evaluation methods***. It provides a consistent and simple API for building end-to-end machine learning pipelines, making it ideal for academic and industrial applications.

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Part A: Dataset Loading and Understanding

1. Install required libraries:

```
pip install pandas scikit-learn matplotlib seaborn
```

2. Import required libraries:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, classification_report
```

3. Load the CSV dataset:

```
df = pd.read_csv("dataset.csv")
```

4. Display first 5 rows of the dataset:

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5. Display dataset shape:

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6. Identify feature columns and target variable:

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7. Check missing values:

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8. Verify class distribution (if classification):

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9. Write initial observations:

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1. Data Preprocessing

Data preprocessing ensures that raw data is transformed into a suitable format for machine learning algorithms.

1. Handle missing values using mean/median/mode:

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2. Encode categorical features:

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3. Apply feature scaling using StandardScaler:

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4. Verify transformed data:

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5. Justify preprocessing choices:

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6. Check for data leakage:

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7. Save preprocessed dataset:

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8. Compare raw vs processed data:

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9. Write preprocessing observations:

2. Train-Test Split

Splitting data helps in evaluating model generalization.

1. Split dataset into training and testing sets:

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2. Use `random_state` for reproducibility:

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3. Verify shapes of train and test data:

3. Model Training

This section focuses on training machine learning models.

1. Train Logistic Regression

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2. Fit models on training data:

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3. Predict on test data:

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4. Compare model outputs:

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5. Store predictions:

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4. Model Evaluation

Evaluation measures model performance using appropriate metrics.

1. Compute accuracy score:

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2. Generate classification report:

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3. Plot confusion matrix:

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4. Interpret precision, recall, and F1-score:

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